

Conor Glackin

Data Scientist | Machine Learning | Computational Biology
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Data scientist with 6+ years' experience developing statistical and machine-learning solutions across biological research and regulated pharmaceutical manufacturing. Completed a PhD in statistics-driven environmental microbiology with strong foundations in mathematics, computational biology and high-dimensional data analysis. Experienced building reproducible workflows, validating models, developing analytical tools and communicating complex findings to scientific, technical and non-technical stakeholders.

Technical Skills

- **Programming:** R, Python, SQL, JMP/JSL, Bash, Git/GitHub, Linux
- **Machine learning and statistics:** Random Forest, LSTM/TensorFlow/Keras, regression, time-series analysis, design of experiments, cross-validation, ROC-AUC and AUPRC
- **Scientific data:** Microbiome, environmental and genomic datasets; NGS, RNA-seq, ChIP-seq, BWA, GATK, SAMtools and ANNOVAR
- **Data and delivery:** Data cleaning, reproducible pipelines, model validation, visualisation, R Shiny, Power BI, ggplot2 and technical reporting

Work Experience

PhD researcher, Leibniz Institute for Baltic Sea Research, Rostock, Germany **Feb. 2022 – Feb. 2026**

- Developed and validated an early-warning system for pathogenic bacterial presence using Random Forest, time-lagged models and LSTM across ~1,500 samples and more than 2,000 variables.
- Built reproducible R and Python/TensorFlow pipelines for multi-source data integration, feature preparation, model training, evaluation and visualisation.
- Designed spatial and temporal splits to test model generalisation across unseen locations and time periods, addressing severe class imbalance using ROC-AUC and AUPRC.
- Translated modelling results into a practical public-health forecasting framework and presented findings at multiple international conferences.
- First author of “**AI-Driven Forecasting of *Vibrio vulnificus* in the Southern Baltic Sea Using High-Resolution Data,**” published in Water Research; co-authored a machine-learning benchmark across 13 biological datasets.

Data Scientist, Abbott Laboratories, Donegal, Ireland **Nov. 2019 - Jan. 2022**

- Delivered statistical analysis, validation activities, documentation, sampling plans, root-cause investigations, SQL-supported reporting workflows, Power BI dashboards, and process optimisation projects in a regulated pharmaceutical manufacturing environment
- Conducted **comparative statistical analysis of clinical (patients' blood) and synthetic data** for glucose conversion factors, which was used for blood glucose testing during manufacturing
- Led development of the Smart New Analytics Project, an **internal user-facing analytics interface** used across multiple teams for reviewing data and reports, building automated reproducible data pipelines and data interfaces
- Led €30,000 **SQL data warehouse upgrade**, collaborating with the internal team and external vendors to implement improved data infrastructure
- **Automated** reporting workflows producing graphs and reports from product testing data (Excel files), reducing manual processing time from **8 hours to 2 minutes**
- Led **Lean Six Sigma Process Improvement project on predictive modelling** for on-site blood glucose strip testing using upstream variables as predictors

Mathematics and Statistics Teaching, University of Galway **Sep. 2018 – Jul. 2019**

- Taught undergraduate students in mathematics, applied mathematics, statistics, and probability.

Education

PhD, **Statistics in Environmental Microbiology,**

Leibniz Institute for Baltic Sea Research, Rostock, Germany.

Feb. 2022 – Feb. 2026

MSc, **Computational Genomics,** NUI Galway, Ireland.

Sep. 2018 – Aug. 2019

BSc, **Mathematics,** NUI Galway, Ireland.

Sep. 2014 – May. 2018

Additional Certifications & Skills

- Lean Six Sigma Yellow Belt [Abbott Diabetes Care, 2021] | Advanced statistical software training (JMP)